	ADITYA COLLEGE OF ENGINEERING AND **No 2/9, Kamakshipura, Hesaraghatta Hobli, Yelahanka, Bengaluru-** Affiliated to VTU, Belagavi | Approved by UGC/AICTE/Govt. of Bengaluru **DEPARTMENT OF MATHEMATICS**	Format No.	ACD-ASS
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ASSIGNMENT-1

Course Name: Discrete Mathematical Structures

Subject Code : BCS305A

Branch/Sem: III SEM All sections

Scheme: 2022 CBCS

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SL NO	QUESTIONS	CO	BTL	Marks
1	Prove the following logical Equivalence without using truth tables i) $[(p \to q) \wedge \{\neg q \wedge (r \vee \neg q)\}] \Leftrightarrow \neg(q \vee p)$ ii) $\{[\neg p \wedge (\neg q \wedge r)] \vee (q \wedge r) \vee (p \wedge r)\} \Leftrightarrow r$	CO1	L3	6M
2	Prove the following using laws of logic $p \to (q \to r) \Leftrightarrow (p \wedge q) \to r$	CO1	L3	6M
3	Establish the validity of the following argument using rules of inference $\{(p \to q) \wedge (r \to s) \wedge (\neg q \vee \neg s)\} \to \neg(p \wedge r)$	CO1	L3	6M
4	Find whether the following argument is valid: If a triangle has two equal sides, then it is isosceles If a triangle is isosceles, then it has two equal angles. A certain triangle PQR does not have two equal angles $\therefore$ The triangle PQR does not have two equal sides.	CO1	L3	7M
5	Prove the following is a valid argument $p \to (q \wedge r)$ $r \to s$ $\neg(q \wedge s)$ $\therefore \neg p$	CO1	L3	7M
6	Write the following in symbolic form and find its negation "If all triangles are right angled then no triangle is equilateral"	CO1	L3	7M

7	For Inverse of all real numbers, let $p(x): x \geq 0$, $q(x): x^2 \geq 0$, $r(x): x^2 - 3x - 4 = 0$, $s(x): x^2 - 3 > 0$. Determine the truth values of the following: i) $\exists x, p(x) \wedge q(x)$ ii) $\forall x, p(x) \rightarrow q(x)$ iii) $\forall x, q(x) \rightarrow s(x)$ iv) $\forall x, r(x) \vee s(x)$	CO1	L3	7M
8	Give i) direct proof ii) indirect proof iii) proof by contradiction for the following statement; " If n is odd integer then n+9 is an even integer".	CO1	L3	7M
9	Give an indirect proof of the statement; " The product of two even integers is an even integer". Provide proof by contradiction of the following statement:" For every integer n, if n^2 is odd then n is odd".	CO1	L3	7M
10	By mathematical induction prove that for all integers $n \geq 1$ $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \ldots\ldots\ldots + n(n+1) = \frac{1}{3} n(n+1)(n+2)$	CO2	L3	6M
11	By mathematical induction prove that $(n!) \geq 2^{n-1}$ for all integers $n \geq 1$	CO2	L3	6M
12	Find an in explicit form for the following i) $a_1 = 4$, $a_n = a_{n-1}$ for $n \geq 2$ ii) $a_1 = 7$, $a_n = 2a_{n-1}$ for $n \geq 2$ iii)	CO2	L3	6M
13	A sequence $\{a_n\}$ is defined recursively by $a_0 = 1$ $a_1 = 1$ $a_2 = 1$ and $a_n = a_{n-1} + a_{n-3}$ for all $n \geq 3$. Prove that $a_{n+2} \geq \left(\sqrt{2}\right)^n$ for all integers $n \geq 0$	CO2	L3	6M
14	How many words can be made with or without meaning from the letters of the word "STATISTICS"? In how many of these a and c are adjacent? In how many vowels are together.	CO2	L3	7M
15	In how many ways of arrangement of the letters of the word 'TALLAHASSEE" which have no adjacent A's.	CO2	L3	7M
16	In how many ways can we distribute 7 apples and 5 oranges among 3 children such that each child gets atleast one apple and one orange	CO2	L3	7M
17	From seven consonants and five vowels, how many sets consisting of four different consonants and three different vowels can be formed?	CO2	L3	7M

18	A woman has 11 close relatives and wishes to invite 5 of them for dinner. In how many ways can she invite them if i) there is no restriction on her choice ii) 2 persons will not attend separately iii) 2 persons will not attend together.	CO2	L3	7M
19	In how many ways can 10 identical pencils be distributed among 5 children in the following case: i) There are no restrictions ii) Each child gets atleast one pencil iii) The youngest child gets atleast 2 pencils	CO2	L3	7M
20	Find the coefficient of $a^2\ b^3\ c^2\ d^5$ in the expansion of $(a+2b-3c+2d+5)^{16}$	CO2	L3	7M
21	Determine the coefficient of i) xyz^2 in the expansion of $(2x-y-z)^4$ ii) $x^2\ y^2\ z^3$ in the expansion of $(3x-2y-4z)^7$	CO2	L3	7M